

Maintenance, Inspection and Troubleshooting

Stamping press 01 (press-stamp-01) | Main drive and lubrication circuit

Machine	Stamping press 01 (machine ID press-stamp-01, type press, area Press shop, site vit-lab), stamping press in a car plant
Sensors	MAIN_MOTOR_CURRENT (A), LUBE_OIL_PRESSURE (bar), BEARING_VIBRATION_RMS (mm/s) and the 64-bin spectrum BEARING_SPECTRUM (mm/s); sensor IDs PRESS-STAMP-01.<CHANNEL_KEY>
Components / subsystem	Press main drive and its pressure lubrication circuit: main drive motor, lubrication unit (gear pump, reservoir, lube filter) and main bearing housing; planned agent tags main_drive_motor, lube_filter, lubrication_unit, main_bearing_housing
Document title	Maintenance, Inspection and Troubleshooting
Document ID	PS01-DOC-04
Revision / date	1.0 / 2026-10-08
Purpose	Translates abnormal behaviour of press-stamp-01 (main drive motor current, lube oil pressure, bearing vibration) into practical inspection, troubleshooting, corrective-action, verification and return-to-service steps for the lubrication circuit, the main bearings and the main drive.
Scope	Condition-based maintenance for the main drive and its pressure lubrication circuit and for the data path of the three channels. The repository holds only two runbook stubs; everything else here is engineering reference [E] to be replaced or confirmed by the OEM manual. Levels are in Document 01, signatures in Document 02, decisions in Document 03, cases in Document 05.

Provenance tags used throughout this document

[P]	Project value: taken from the repository or CONTEXT_PRESS_STAMP_01.md (catalog limits, simulation equations, topics, runbook stubs, tests).
[D]	Derived value: computed from the repository's own simulation model (seed 1, 2 Hz, 600 s per case) or algebraically from its equations, as recorded in the context file.
[E]	Engineering reference: general industrial practice for press drives, lubrication circuits and vibration diagnostics. Not a manufacturer value and not taken from the repository.
[I]	Illustrative / proposed prototype value: chosen by the authors for this knowledge base and meant to be refined with real plant data.

Important: the telemetry for this machine is simulated. There is no physical press or transducer. An ESP32 gateway (or a software simulator) plays the press controller and generates values from documented equations. The repository describes its limits as illustrative constants chosen for the demo, not vendor values, and its runbooks as not an OEM manual. Values tagged [P] are therefore prototype values, not OEM specifications; values tagged [E] and [I] are not manufacturer-certified.

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1. Maintenance Philosophy

1.1 Condition-based approach

The repository gives no inspection or service intervals, filter or bearing part numbers, oil specification, differential-pressure limits, torque values, replacement procedures, tools, spares, mean time between failures or OEM references [P]. This document therefore does not invent intervals. Work is triggered by condition: the Document 01 level (L1 to L4, LS), the diagnosis and the verification result. Routine inspection follows the OEM schedule and the site plan, which are referenced but not reproduced.

PRINCIPLE (press runbooks [P]): do not run a drifting bearing to alarm; schedule an inspection at the next planned stop. The cheap filter fix is only cheap while it is early.

Table 1. Condition-based maintenance triggers

Telemetry condition	Level	Maintenance response	Timing basis
All features inside healthy bands	L0	Routine inspection per OEM schedule	OEM schedule (not specified in repository)
V60 at or above 1.50 mm/s or P60 at or below 4.30 bar, no limit	L1	Add to next planned inspection; check filter and oil records; trend	Next planned inspection
V or P at warn (2.5 mm/s or 3.6 bar)	L2	Filter inspection; bearing inspection if lines present	Next planned stop
V at 4.0 mm/s or P at 3.0 bar	L3	Filter swap or intervention; plan bearing check	This shift [I]
P at or below 2.5 bar, V at or above 7.0 mm/s, lpk at or above 220 A	L4	Controlled stop and full inspection	Immediate
Flat, implausible or stale data	LS	Sensor and data-path inspection first	Within the shift

1.2 What the repository does and does not provide

Table 2. Maintenance information availability

Topic	Status	Treatment here
Filter replacement as the action for a clogged filter, minutes of work, acceptance values	Available (lubrication runbook)	Used as stated [P]
Bearing wear action: inspect at next stop, confirm defect frequencies with a handheld analyser, plan replacement	Available (bearing runbook)	Used as stated [P]
Safety wording: manufacturer's procedure, lock-out and tag-out, relieve line pressure, specified cartridge, specified bearings and fits	Available	Applied in Section 2
Service intervals, part numbers, oil grade, torque values, step procedures, tools, spares	Not specified in repository	Condition-based; OEM values required [E]
Data-path troubleshooting table	Available (firmware README, contract)	Reproduced in Section 21
Maintenance history	Not specified in repository	Cases in Document 05 are simulated

2. Safety Considerations

This document is a capstone prototype. It does not replace the OEM manual, the plant lock-out and tag-out procedure or site safety rules. Follow the manufacturer's procedure. Apply lock-out and tag-out before opening the lubrication circuit or the main drive and relieve the line pressure. Only qualified persons may do the work.

Table 3. Hazards and controls [E]

Hazard	Where	Control
Stored energy in the slide, flywheel and clutch	Press	Isolate and block the slide per OEM; verify zero energy
Pressurised oil	Lube circuit downstream of the pump	Lock-out; relieve line pressure; open slowly
Hot oil and surfaces	Reservoir, housing, motor	Allow cooling; gloves, eye protection
Unexpected start	Main drive	Lock-out and tag-out; personal locks; try-out test
Electrical shock, stored charge	Motor, drive	Qualified electrician; verified isolation; rated meters
Rotating parts	Shaft, coupling, flywheel	Guards in place; no work while rotating except approved vibration measurement
Crush at the die	Ram and die	Die blocks, safety props
Oil spill and slips	Floor	Absorbents; clean up; safety data sheets
Manual handling	Cartridge, bearings	Lifting aids
Restart after work	Controller	Verification run per Section 14; clear the area

2.1 Pre-work checklist

- Confirm the machine ID (press-stamp-01), the level and the diagnosis from the report.
- Obtain the OEM manual, the permit to work and the lock-out and tag-out plan.
- Record baseline readings before work: P60, V60, I60, HB300 and a spectrum snapshot.
- Identify the specified cartridge, bearings and oil from OEM documents, not from this document.
- Check that no cmd/scenario test is running (a simulated fault must not be mistaken for a plant fault).

3. Routine Inspection Checklist

Interval column states the basis, not a number. A telemetry trigger can advance any item.

Table 4. Routine inspection checklist

Item	What to check	Interval basis	Telemetry trigger that advances it	Record
Lube filter	Condition, differential pressure if a gauge exists, cartridge date	OEM schedule; condition	P60 at or below 4.30 bar; HB300 rising	Date, condition
Oil level and condition	Level, colour, water, particles	OEM schedule	Pressure erratic	Level, sample ID
Pump and drive coupling	Noise, leaks, coupling	OEM schedule	Pulsation changed	Condition
Lines, fittings, nozzles	Leaks, kinks, blockage	OEM schedule	P rising or falling	Condition
Pressure gauge or reference port	Compare to LUBE_OIL_PRESSURE	Each inspection	P differs from the gauge	Both readings
Main bearing housing	Temperature by hand or instrument, noise, mounting	OEM schedule	V60 above 1.50 mm/s	Condition
Bearing vibration check	Handheld spectrum at the housing	Each planned stop where available; condition	Lines at defect orders	Spectrum file
Main drive motor	Cooling, terminals, noise	OEM schedule	I60 above 99.5 A	Condition
Foundation and fasteners	Bolts, grout, cracks	OEM schedule	Integer harmonics	Torque check
Gateway and data path	status online, synced true, no seq gaps, diag fields	Weekly for the prototype [I]	stale or gaps	diag snapshot
Baseline	Recompute features	After any repair, die change or oil change	Any L1	Baseline record

3.1 Inspection round sheet

Table 5. Inspection round sheet (fields to complete) [E]

Field	Entry
Machine / date / inspector	press-stamp-01 / date / name
Report ID and level	From the diagnosis report
Baseline before work	P60, V60, I60, HB300
Filter condition grade	A, B, C or D
Oil level and condition	Level, colour, particles
Handheld spectrum	File name; lines present or absent
Sensor agreement	Gauge, analyser and clamp meter readings
Findings and actions	Text
Verification result	Pass or fail with values

4. Mechanical Inspection

4.1 Main bearing housing

Table 6. Main bearing housing inspection criteria [E]

Check	Method	Acceptable	Action if not
Defect lines	Handheld analyser, spectrum to 800 Hz	No lines at about 88, 175, 263, 351, 439 Hz with sidebands at 24.7 Hz	Plan bearing replacement [P]
Overall vibration	Handheld RMS at the housing	Agrees with BEARING_VIBRATION_RMS within the sensor tolerance	Check sensor
Housing temperature	Touch or contact thermometer	No local hot spot	Check lubrication
Noise	Listen	No rumble or clicking	Analyser
Mounting and fasteners	Visual, torque check per OEM	Marks intact	Tighten per OEM
Oil film evidence	Oil at the bearing drain	Clean oil, no metal	Check filter first

4.2 Handheld vibration measurement procedure

Table 7. Handheld vibration measurement [E]

Step	Action	Pitfall to avoid
1	Measure with the press running steadily, not during a die change	Transient state gives unrepresentative amplitudes
2	Place the probe at the housing in the same position and direction as the installed sensor	Different position changes amplitude
3	Set the span to at least 800 Hz and a resolution that separates 24.7 Hz sidebands	Too coarse a resolution merges lines
4	Record the overall RMS and the spectrum	Not recording the file
5	Mark 1X (24.7 Hz) and the defect orders 3.55, 7.1, 10.65, 14.2, 17.75 x 1X [P]	Confusing integer orders with defect orders
6	Compare with the channel value and with the healthy signature	Comparing across different operating states

4.3 Main drive and shaft

Table 8. Main drive inspection criteria [E]

Check	Method	Acceptable	Action if not
1X amplitude	Handheld spectrum	About 1.05 mm/s on the sensor channel baseline	Alignment or balance
Coupling and alignment	Visual, dial or laser per OEM	Within OEM	Re-align
Flywheel and guard	Visual	No damage	Repair
Belts or gears if fitted	Visual	Not specified in repository	Per OEM
Backlash and wear	Not specified in repository	OEM	Per OEM

5. Lubrication Circuit Inspection

5.1 Circuit elements

Table 9. Lubrication circuit inspection

Element	Inspection	Acceptable	Telemetry link
Reservoir	Level, temperature, condition	Between marks; clean	None (level not measured [P])
Gear pump	Noise, coupling, delivery	Smooth; pressure about 4.5 bar with 0.12 bar pulsation	Pulsation in LUBE_OIL_PRESSURE
Lube filter	Cartridge condition, delta-P if gauge	Clean; no bypass	P60 falling with V60 rising when clogged
Pressure transducer	Reading against a reference gauge	Agrees	LUBE_OIL_PRESSURE
Distribution line and nozzles	Leaks, blockage	Free flow	P rising when blocked after the transducer
Relief valve	Setting, sticking	Within OEM	Pressure step

5.2 Oil sampling and analysis

Table 10. Oil sampling [E]

Item	Method	Interpretation
Sample point	Per OEM, from the circuit while running	Representative sample
Particle count	Laboratory	High counts support a clogging filter (F01)
Metals	Laboratory	Iron or bearing alloys support bearing wear (F02)
Water	Laboratory	Degradation and corrosion
Viscosity	Laboratory	Affects pressure (F07)
Interval	Not specified in repository	Take a sample when a lubrication condition is suspected

5.3 Interpreting what the filter looks like

Table 11. Filter condition assessment [E]

Observation	Grade	Meaning	Action
Clean media, normal delta-P	A	Healthy	None
Light loading	B	Early loading	Monitor
Heavy loading, discoloured media	C	Clogging	Replace
Collapsed media, bypass open	D	Failed	Replace now; inspect bearings
Metal particles in media	D+	Wear debris	Replace and inspect bearings and pump

Metal particles in a clogged filter point beyond the filter to bearing or pump wear. Replace the cartridge and inspect the main bearings before returning to service [E].

6. Electrical Inspection

Table 12. Electrical inspection criteria [E]

Check	Method	Acceptable	Link to telemetry
Motor terminals and cables	Visual, torque per OEM	Tight, no heat marks	I rises or noisy
Insulation resistance	Insulation tester per OEM	Within OEM limits	I above 8 percent with line-frequency lines
Supply balance	Meter	Within OEM tolerance	I changes for the same load
Drive parameters and trips	Drive display and log	No faults	lpk above 175 A
Current measurement chain	Clamp meter versus MAIN_MOTOR_CURRENT	Within sensor tolerance	Flat or offset I
Grounding and shielding of sensor cables	Visual, continuity	Intact	Spikes in V

7. Sensor Inspection and Validation

Table 13. Sensor inspection by channel [E]

Channel	What to inspect	Reference	Acceptance	If it fails
LUBE_OIL_PRESSURE (0 to 8 bar, 2 decimals)	Port, fitting, wiring, ripple signature	Reference gauge at the same port	Agrees within the transducer tolerance; ripple 0.12 bar and noise 0.03 bar visible	Replace or recalibrate
BEARING_VIBRATION_RMS (0 to 12 mm/s) and BEARING_SPECTRUM	Mount torque, cable, shielding	Handheld analyser at the same housing	RMS agrees; spectrum has 1X as largest bin; no mains lines	Refit, replace cable
MAIN_MOTOR_CURRENT (0 to 250 A)	Burden, wiring	Clamp meter	Agrees at idle (about 88 A) and in stroke (about 150 A)	Replace or recalibrate
Gateway time	synced true	SNTP	ts within 2 minutes	Fix NTP (UDP 123)

Before touching the filter or the bearings, confirm that the symptom is not a sensor symptom: a flat signal, a missing ripple, a mismatch between RMS and spectrum or a change not seen on any other channel are sensor and data signatures.

8. Component Condition Assessment

Table 14. Condition grades for the three components [E]

Component	Grade A (good)	Grade B (watch)	Grade C (act)	Grade D (replace now)
Lube filter	Clean; P 4.4 to 4.6 bar	Light loading; P60 4.30 bar or lower	Heavy loading; P60 at or below 3.6 bar	Bypass or collapse; P at or below 3.0 bar
Main bearings	V60 1.2 to 1.45 mm/s; no lines	V60 above 1.50; faint lines	Lines at least 4 x healthy; V60 above 2.0	V60 above 2.5 mm/s with strong lines, noise or heat
Main drive motor	I60 within 3 percent of baseline	I60 above 1.05 x	IpK above 175 A or heating	Trips, insulation failure
Lubrication unit (pump, relief valve)	P 4.5 bar with ripple	Ripple changed	P step or irregular	No pressure
Sensors	Agree with references	Drift within tolerance	Out of tolerance	Failed

Table 15. Grade to action

Grade	Action	Timing
A	Routine	OEM schedule
B	Add to next planned inspection	Next planned inspection
C	Plan the work and parts	Next planned stop
D	Replace	This shift or immediately

8.1 Interpreting inspection findings against telemetry

Table 16. Inspection finding versus expected telemetry [E]

Finding on inspection	Expected telemetry	If telemetry disagrees
Heavily loaded filter	P60 low, V60 up, HB300 up	Suspect transducer or a second cause
Clean filter	P60 normal	Pressure fall has another cause: pump, leak, valve, transducer
Metal in the filter or oil	V60 up with lines at defect orders; P may be normal	Sample oil; inspect pump too
Bearing noise or heat	V60 up; I60 up	Check alignment and lubrication
Loose fasteners	Integer harmonics of 1X	Tighten and re-measure
Worn coupling or misalignment	1X and 2X up	Align
Damaged vibration cable	V spikes or mismatch	Replace and re-baseline

9. Troubleshooting Procedure

9.1 Procedure steps

Table 17. Troubleshooting procedure

Step	Action	Decision
1	Confirm the report: channel, value, window, level	If the data are not valid, go to Section 21
2	Rule out the data path and sensors (Sections 7 and 21)	If a sensor or link is faulty, repair it first
3	Confirm the pressure trend over the last hour and the pump pulsation	Falling: lubrication path; normal: bearing or other
4	Confirm the spectrum shape: broadband or peaked	Broadband above 300 Hz: filter; peaked at defect orders: bearing
5	Confirm the other channels are normal	If current is the only abnormal channel, go to the process and drive checks
6	If the evidence is mixed, check the lubrication first	Replace the filter, then re-check the spectrum
7	Apply lock-out and tag-out, perform the inspection or repair	Record findings
8	Verify against the criteria of Section 14	Return to service or reopen

9.2 Troubleshooting decision tree (text)

Table 18. Decision tree

Question	Yes	No
Is the data valid (online, no gaps, RMS equals spectrum)?	Next question	Section 21; repair data path
Is the press running steadily?	Next question	Wait; suppress diagnosis
Is P60 at or below 4.30 bar?	Check the spectrum floor; go to filter	Next question
Are there discrete lines at defect orders?	Bearing inspection T2	Next question
Is Ipk at or above 160 A with idle 88 A?	Process and drive T8	Next question
Is any channel flat or implausible?	Sensor T4 to T6	Trend and watch

9.3 Observed condition to action

Table 19. Observed condition, possible cause, inspection, corrective action, verification

Observed condition	Possible cause	Inspection	Corrective action	Verification
P60 at or below 3.6 bar, V60 at or above 2.5 mm/s, flat floor	Clogged filter (F01)	Cartridge, delta-P	Replace cartridge	P60 4.5 bar, V60 1.3 mm/s
P60 at or below 3.0 bar, V60 above 4.0 mm/s	Severe clog or starved bearing	Cartridge, then bearings and oil	Replace cartridge now; plan bearing check	P and V back; lines absent
V60 rising, P normal, lines at defect orders	Bearing wear (F02)	Handheld analyser, housing	Plan bearing replacement	Lines gone, V60 1.3 mm/s
V60 at 2.5 to 3.5 mm/s, P normal, no lines	Imbalance, looseness, sensor	Alignment, fasteners, mount	Correct	V60 1.3 mm/s
P low, V normal	Transducer, pump, leak, relief valve	Reference gauge, level, pump	Repair	P 4.5 bar with ripple
P above 4.65 bar, rising	Blocked nozzle or line after the transducer	Nozzles, relief valve	Clear restriction	P 4.5 bar
l _{pk} at or above 175 A, idle 88 A	Process or die load	Die, material, shut height	Correct setup	l _{pk} about 150 A
I60 above 8 percent, line-frequency lines	Motor or supply	Insulation, supply	Repair motor	I60 94.8 A
I flat or near zero, press cycling	Current sensor	Clamp meter	Repair sensor	Pulse to about 150 A
V10 at or above 7.0 mm/s, P and I normal	Mount or cable	Mount, shield	Refit or replace	V about 1.3 mm/s
RMS differs from spectrum	Decoding or data	Gateway, backend	Fix data path	RMS equals spectrum
Pressure drops but filter swap does not help	Pump, leak, relief valve, transducer	Level, pump, gauge	Repair	P 4.5 bar
V rises again after a bearing change	Lubrication or alignment	Filter, alignment	Correct	V 1.3 mm/s
Alarm appears only at stroke peaks	Per-sample classification	Window means	No repair	V10 inside limits
Pressure alarm after cold start	Cold oil	Oil temperature	Warm up	P 4.5 bar warm
Device stale	Network or power	diag, status	Fix link	status online
Identity conflict flagged	Board and simulator both announce press-stamp-01	Stop one source	Stop the simulator or the board	Single source
Fault seems stuck	Simulated fault active	Faults tab	Choose Normal or send cmd/reboot	Values normal

10. Fault-to-Action Matrix

Table 20. Fault-to-action matrix

Fault	Immediate action	Planned action	Parts and tools (OEM-specified)	Verification
F01 CLOGGED_FILTER	Replace cartridge (minutes)	Investigate oil cleanliness	Specified cartridge	P60 4.5 bar, V60 1.3 mm/s
F02 BEARING_WEAR	Do not run to alarm	Inspect at next stop; plan replacement	Specified bearings and fits; handheld analyser	Lines absent, V60 1.3 mm/s
F03 pump wear	Controlled stop if P at or below 3.0 bar	Repair or replace pump	OEM pump	Ripple and P restored
F04 leak or low level	Refill, find leak	Repair	Specified oil, seals	P stable
F05 blockage after transducer	Reduce load if needed	Clear line or nozzle	OEM tools	P 4.5 bar
F06 relief valve	Check setting	Service valve	OEM kit	P restored
F08 motor or supply	Stop if trip or heat	Test and repair	Electrical test set	I60 94.8 A
F09 process	Correct setup	Review die	Die tools	IpK about 150 A
F10, F11	Re-check fasteners	Align, balance	Alignment tools	1X back to about 1.05 mm/s
F12 to F14	Verify against reference	Repair sensor	Reference instruments	Signals agree
F15	Restore link	Root-cause network	Network tools	No gaps
F16	Replace filter first	Inspect and plan bearing replacement	As F01 and F02	Both verified

10.1 Planning the work from the estimated severity

Table 21. Work planning from the estimated severity [D] / [I]

Estimated severity (r equivalent)	Clog signature (P60, V60)	Bearing signature (V60)	Planning
0 to 0.1	4.50 to 4.30 bar; 1.31 to 1.68 mm/s	1.31 to 1.47 mm/s	Routine; baseline check
0.1 to 0.3	4.30 to 3.89 bar; 1.7 to 2.4 mm/s	1.47 to 1.78 mm/s	Plan inspection; order cartridge if clog
0.3 to 0.5	3.89 to 3.49 bar; 2.4 to 3.15 mm/s	1.78 to 2.10 mm/s	Next planned stop
0.5 to 0.75	3.49 to 3.0 bar; 3.15 to 4.0 mm/s	2.10 to 2.50 mm/s	This shift (clog); plan bearing change
0.75 to 1.0	3.0 to 2.47 bar; 4.0 to 5.0 mm/s	2.50 to 2.89 mm/s	Immediate action

11. Corrective Priorities and Replacement Criteria

Table 22. Corrective priority

Priority	Condition	Reason
1	LS: sensor or data fault	Avoids unnecessary mechanical work
2	L4 or P at or below 2.5 bar	Oil starvation
3	L3 with broadband floor	Cheap, high-benefit filter swap
4	L3 or L2 with lines (bearing)	Planned replacement; avoid failure in service
5	L2 process or electrical	Planned correction
6	L1	Trend and inspection

Table 23. Replacement criteria [E]

Component	Replace when	Do not replace when
Lube filter cartridge	Loaded or bypassing (grade C or D), or P60 at or below 3.6 bar with broadband floor confirmed	Pressure low but spectrum peaked, or sensor not verified
Main bearing	Lines at defect orders at least 4 x healthy with rising V60, or metal in the oil, or noise or heat, or OEM limit reached	Single spike; sensor fault; integer-order pattern only (alignment first)
Pressure transducer	Disagrees with the reference gauge beyond tolerance	Pressure fall confirmed by V60 and HB300
Vibration sensor	Disagrees with the handheld analyser or RMS differs from the spectrum	Vibration confirmed by handheld
Gear pump	No delivery or pulsation lost, filter excluded	Filter loaded
Oil	Contaminated or degraded per analysis	Not specified in repository: no interval given

Replace parts only with the OEM-specified cartridge, bearings and fits [P]. The repository gives no part numbers.

12. Task Procedures

Task procedures are generic engineering reference [E]. The OEM procedure takes precedence. Steps that are the repository's own statements are marked [P].

12.1 T1 Replace the lube filter cartridge (F01) [P]

Table 24. T1: steps

Step	Action	Check / record
1	Confirm diagnosis: P60 falling, flat broadband floor, V60 rising	Record P60, V60, HB300, spectrum snapshot
2	Obtain permit; apply lock-out and tag-out; relieve line pressure [P]	Zero-energy verified
3	Open the filter housing per OEM; remove the cartridge	Inspect media: grade; metal particles?
4	Fit the specified cartridge [P]; reseal and close the housing	Cartridge ID recorded
5	Refill or bleed per OEM; remove locks; restore power	Area clear
6	Start; watch LUBE_OIL_PRESSURE rise	P about 4.5 bar [P]
7	Run the verification of Section 14	V60 about 1.3 mm/s [P]

12.2 T2 Bearing inspection with a handheld analyser (F02) [P]

Table 25. T2: steps

Step	Action	Check / record
1	At the next planned stop, take the baseline from the sensor channel	Record V60 and spectrum
2	Measure at the housing in the same direction and position as the sensor	Compare RMS
3	Look for lines at about 88, 175, 263, 351, 439 Hz with sidebands at 24.7 Hz	Mark presence and amplitude
4	If lines present: plan bearing replacement; if absent: check alignment and sensor	Decision recorded
5	Do not run to alarm [P]	Schedule and parts per OEM

12.3 T3 Replace the main bearings [P][E]

Table 26. T3: steps

Step	Action	Check / record
1	Plan the stop; obtain bearings and fits specified by the OEM	Parts verified
2	Lock-out and tag-out before opening the main drive [P]	Zero-energy verified
3	Open and remove per OEM; inspect shaft and housing	Wear, damage recorded
4	Fit the specified bearings and fits [P]; reassemble; lubricate per OEM	Fit values recorded
5	Align and torque per OEM	Values recorded
6	Run-in at reduced speed and verify (Section 14)	V60 about 1.3 mm/s, lines absent

12.4 T4 Check the pressure transducer (F12)

Table 27. T4: steps

Step	Action	Check / record
1	Fit the reference gauge at the same port	Both readings
2	Compare under steady running	Agreement within tolerance
3	If different, inspect wiring and port	Condition
4	Replace or recalibrate; repeat	Ripple 0.12 bar visible

12.5 T5 Check the vibration sensor (F13)

Table 28. T5: steps

Step	Action	Check / record
1	Inspect mount and cable	Condition
2	Measure with a handheld analyser at the same position	RMS and spectrum
3	Compare with the channel and with the spectrum root-sum-square	RMSchk
4	Refit or replace	Agreement

12.6 T6 Check the current measurement (F14)

Table 29. T6: steps

Step	Action	Check / record
1	Clamp meter on the motor lead	Idle and stroke values
2	Compare with MAIN_MOTOR_CURRENT	About 88 A idle, about 150 A peak
3	Inspect wiring and burden	Condition
4	Repair; repeat	Match

12.7 T7 Check pump and level (F03, F04)

Table 30. T7: steps

Step	Action	Check / record
1	Check reservoir level and condition	Level
2	Inspect for leaks along the circuit	Leak location
3	Check pump noise and coupling	Condition
4	Run delivery test per OEM	Pressure and ripple

12.8 T8 Check process and electrical causes (F08, F09)

Table 31. T8: steps

Step	Action	Check / record
1	Compare idle current with 88 A	Idle level
2	Compare stroke peak with the die record	Peak value
3	Check supply balance and insulation	Values
4	Correct setup or repair motor	Ipk about 150 A

13. Calibration and Validation

Table 32. Calibration and validation [E]

Item	Method	Frequency basis	Acceptance
LUBE_OIL_PRESSURE	Compare with a calibrated reference gauge at the same port	After sensor replacement; when a pressure alarm lacks corroboration	Within the transducer tolerance (not specified in repository)
BEARING_VIBRATION_RMS	Compare with a handheld analyser	After remounting; when RMS and spectrum disagree	RMS agrees; 1X is the largest bin
MAIN_MOTOR_CURRENT	Compare with a clamp meter	After replacement or if offset suspected	Idle about 88 A; stroke peak about 150 A
Timestamp	Check synced and ts	After network changes	synced true
Baseline	Recompute features	After any repair	Recorded

The repository does not specify sensor accuracy or calibration procedures [P]. The values above are checks of agreement, not calibration certificates.

14. Post-Maintenance Testing

14.1 Test steps

Table 33. Post-maintenance test

Step	Test	Pass criterion	Source
1	Visual and leak check, guards in place	No leaks; guards fitted	[E]
2	Start at reduced speed or per OEM	No abnormal noise	[E]
3	P60 after 5 minutes of running	4.35 to 4.65 bar with ripple	[P] about 4.5 bar
4	V60 after 10 minutes	1.20 to 1.45 mm/s	[P] about 1.3 mm/s
5	Spectrum	1X largest; no lines; HB300 at most 0.06 mm/s	[P] / [I]
6	I60	Within 3 percent of 94.8 A	[I]
7	Steady run	60 minutes without L1 or above	[I]
8	Data	status online, no gaps, RMS equals spectrum	[P]

14.2 Return-to-service criteria

Table 34. Return-to-service criteria

Criterion	Required	Evidence
Safety	Locks removed, guards fitted, area clear	Permit closed
Pressure	Restored within the band	P60
Vibration	Back to baseline	V60, spectrum
Current	Within 3 percent of baseline	I60
Data	Valid for 60 minutes	Data-quality checks
Records	Findings and parts recorded	Work record
Approval	Supervisor sign-off	Signature

Table 35. If a criterion fails

Failed criterion	Meaning	Next step
Pressure not restored	Other lubrication fault or sensor	T4, T7
Vibration not restored after a filter swap	Bearing wear or mixed fault	T2
Vibration not restored after bearing change	Alignment or lubrication	T1, alignment
Current high	Load or friction	T8

14.3 Service events and their effect on telemetry

Table 36. Service events and expected telemetry [E] / [P]

Event	Expected effect	Handling
Filter cartridge replaced	P steps up toward 4.5 bar within the circuit fill time; V settles to about 1.3 mm/s [P]	Compare with the verification band; update baseline
Bearing replaced	V returns to about 1.3 mm/s; lines vanish; I60 returns to about 94.8 A [P] / [D]	Run-in period at reduced speed; re-baseline
Oil changed or topped up	Pressure may change slightly with viscosity	Re-baseline P
Sensor replaced	Step change in that channel only	Re-baseline; check against reference
Die changed	IpK and current baseline change; P and V unchanged	Re-baseline current
Gateway rebooted	bootId changes; seq restarts; simulated fault cleared	Restart window

15. Use of the Digital Twin During Maintenance

Table 37. Twin elements [P]

Element	Behaviour	Use
Mesh labelled Lubrication unit (suggested 090-stamping-press/1)	Amber on warn, pulsing red on alarm	Locate the pressure channel; the label says what it stands for
Mesh labelled Main bearing housing (suggested 090-stamping-press/ram/steel)	Same	Locate vibration
Mesh labelled Main drive motor (suggested 090-stamping-press/0)	Same	Locate current
Telemetry tab	Trend line per channel; spectrum with ticks at the shaft harmonics	Compare to the healthy signature
Faults tab (needs ENABLE_DEVICE_COMMANDS true)	Injects simulated faults	Do not use during real maintenance; choose Normal to reset

The twin asset has no lubrication unit, motor or bearing housing as separate parts; the meshes are suggestions and docs/demo-bindings.json records no captured binding [P]. Keep the pressure and vibration meshes apart and large enough to read, because the fault demo tints both.

16. Tooling and Competence

Table 38. Tools and competence [E]

Task	Tools	Competence
Vibration check	Handheld analyser (to 800 Hz)	Vibration analyst
Filter change	OEM tools, specified cartridge	Mechanical technician
Bearing replacement	OEM tools, specified bearings	Maintenance engineer
Electrical	Insulation tester, meter	Electrician
Sensors	Reference gauge, clamp meter	Instrument technician
Data path	Network tools, broker logs	IT

16.1 Spare parts and consumables

Table 39. Spares and consumables [E]

Item	Basis for stocking	Note
Lube filter cartridge	The only repository-stated action for a clogged filter; keep the OEM-specified type	Part number not specified in repository
Main bearing set and fits	Long lead time; planned replacement	Part number and fits not specified in repository
Lubricant	Per OEM specification	Grade not specified in repository
Pressure transducer, spare	Quick sensor swap	Type not specified in repository
Seals and gaskets	On opening the circuit	OEM kit

17. Operator Symptoms

Table 40. Operator observation to telemetry check

Operator observation	Check	Possible diagnosis
Pressure gauge low (if fitted)	P60, V60, HB300	F01
Bearing noise	V60, spectrum	F02 or F01
Press sounds heavier	l _{pk} , idle level	F09
Oil visible on floor	Level, P	F04
Press stops with drive trip	I	F08
Twin shows nothing	Data path	F15

18. Work Packages by Level

Table 41. Work packages

Level	Package	Content	Duration basis
L1	Inspection	Filter condition, oil, spectrum check, baseline update	Condition-based
L2	Filter and bearing check	T1 if broadband; T2 if lines; sensor check	Next planned stop
L3	Intervention	T1 now; T2 afterwards; verification	This shift
L4	Controlled stop	Full inspection T1, T2, T7, T8	Immediate
LS	Data and sensors	T4 to T6 and Section 21	Within the shift

19. Prevention and Common Mistakes

Table 42. Common mistakes

Mistake	Effect	Correct practice
Replacing the bearing when the filter is clogged	Cost and downtime	Lubrication first [P]
Waiting for a bearing alarm that never comes	Failure in service	Use trend and spectrum [P]
Trusting a flat pressure	Missed fault	Check sensor
Using a single sample	False alarm	Window means
Skipping lock-out and tag-out	Injury	Follow the procedure
Not recomputing the baseline	False alarms	After every repair
Leaving a simulated fault active	False diagnosis	Normal or reboot

20. Escalation and Records

Table 43. Escalation

Situation	Escalate to	When
L3 or L4	Supervisor, manager	Immediately
Bearing replacement needed	Maintenance engineer	Planning
Repeated recurrence	Reliability engineer	After second occurrence
Sensor disagreement	Instrumentation	Same shift
Safety event	Safety officer	Immediately

Table 44. Maintenance record fields

Field	Content
Machine and component	press-stamp-01; filter, bearing, drive
Trigger	Level, features, report ID
Findings	Condition grade, photos, analyser files
Work done	Task ID, parts, OEM references
Verification	Criteria and values
Return to service	Date, approver

21. Data-Path Troubleshooting

Rule out these before a mechanical cause. They come from the repository's firmware README, MQTT contract and demo script [P].

Table 45. Data-path symptoms and likely causes [P]

Symptom	Likely cause
Telemetry arrives but the twin shows nothing	Channel not bound to a mesh, or the machine not added to the twin
Device shown stale	No telemetry for max(5 s, 4 x intervalMs)
Device offline about 15 s after power loss	Last Will published by the broker
Identity conflict flagged	Board and simulator both announcing press-stamp-01
A fault seems stuck	Choose Normal in the Faults tab, or send cmd/reboot
No Faults tab	ENABLE_DEVICE_COMMANDS is not true
wifi disconnect event, reason 201	Network not found, or 5 GHz only
reason 202 or 15	Wrong Wi-Fi password or unsupported security mode
mqtt connect failed (state 4) / state 5	Wrong broker credentials / credential lacks permission
tls error -9984	Clock wrong, or the CA file does not anchor the chain
clock not set after 15 s	Network blocks NTP (UDP 123)
Gaps in seq	Packet loss

Table 46. Gateway health fields to inspect (diag) [P]

Field	Healthy
uptimeS	Increasing
rsi	Stable
heapFree, heapMin	Stable minimum over a 30 minute soak
reset	Expected reason
wifiReconnects, mqttReconnects, restarts, publishFailures	Low and stable
tls	true

22. Glossary

Table 47. Glossary

Term	Meaning
Clogged filter	Lube filter loaded with debris; downstream pressure falls (CLOGGED_FILTER)
Outer-race defect	Bearing defect that produces lines at non-integer orders (BEARING_WEAR)
1X	Shaft frequency 24.7 Hz
HB300	Mean spectrum amplitude above 300 Hz
PMR300	Peak-to-median ratio above 300 Hz
LOTO	Lock-out and tag-out
Baseline	Healthy reference values of the channel
Ground truth	diag.sim, hidden from the agent

23. Summary, Safety, References and Revision

23.1 Maintenance summary

- Condition-based: filter replacement for broadband and falling pressure; planned bearing work for discrete lines; sensors and data first when signals look wrong.
- Mixed evidence: lubrication first [P]. Filter swap is minutes; a bearing that fails in service costs days [P].
- Verify: pressure about 4.5 bar, vibration about 1.3 mm/s, spectrum back to healthy shape [P].

23.2 Key criteria

Table 48. Key criteria

Item	Value	Tag
Verification P60	4.35 to 4.65 bar	[P] / [I]
Verification V60	1.20 to 1.45 mm/s	[P] / [I]
HB300 after repair	At most 0.06 mm/s	[I]
Replace filter at	P60 at or below 3.6 bar with broadband floor	[P] / [E]
Bearing action	Lines at defect orders: plan replacement	[P]

23.3 Safety and escalation notes

This document is a capstone prototype, not an OEM manual. Follow the manufacturer's procedure, apply lock-out and tag-out, relieve line pressure and use the specified cartridge and bearings. The agent cannot act on the machine.

23.4 References and engineering basis

- docs/runbooks/press-lubrication-circuit.md and press-main-bearings-vibration.md [P]; firmware README and MQTT contract for the data path [P].
- General maintenance practice for pressure lubrication circuits and rolling-element bearings [E].

23.5 Revision notes

Table 49. Revision history

Rev	Date	Change
1.0	2026-10-08	First issue